

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour

Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Code of Conduct:

I P.Mokshith Reddy(192425149)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

P. MOKSHITH REDDY

Q1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20

during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Answer:

*The neural network has a **training Mean Squared Error (MSE) of 0.85** but a much higher **testing MSE of 5.20**. This large difference indicates that the model performs very well on the training data but poorly on unseen test data. This is a sign of **overfitting**.*

Analysis:

Training MSE = 0.85: The model has learned the training data very accurately.

Testing MSE = 5.20: The model cannot generalize well to new data, resulting in high prediction errors.

Possible Reasons:

Overfitting: The neural network has memorized the training data instead of learning general patterns.

Insufficient Training Data: A small dataset makes it difficult for the model to generalize.

Complex Model: Too many hidden layers or neurons increase the chance of overfitting.

No Regularization: Lack of techniques such as dropout or L2 regularization allows the model to fit noise.

Data Distribution Difference: Training and testing datasets may have different characteristics

Noisy or Irrelevant Features: Noise in the data can reduce prediction accuracy on the test set.

Corrective Measures:

Use Regularization: Apply **L1/L2 regularization** or **dropout** to reduce overfitting.

Increase Training Data: Use more training samples or apply **data augmentation** where applicable.

Simplify the Model: Reduce the number of layers or neurons.

Apply Early Stopping: Stop training when the validation error starts increasing.

Perform Cross-Validation: Use k-fold cross-validation to improve generalization.

Feature Selection and Data Cleaning: Remove irrelevant features and noisy data.

Hyperparameter Tuning: Optimize learning rate, batch size, number of epochs, and network architecture.

Conclusion:

The significant gap between **training MSE (0.85)** and **testing MSE (5.20)** indicates **overfitting**. By using regularization, early stopping, more training data, model simplification, and proper hyperparameter tuning, the neural network can achieve better generalization and improve its performance on unseen data.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided